# Independent prime-field lead controls

**Two finite cycles · 2026-11-09 · terminal snapshot 05:55:17 UTC**

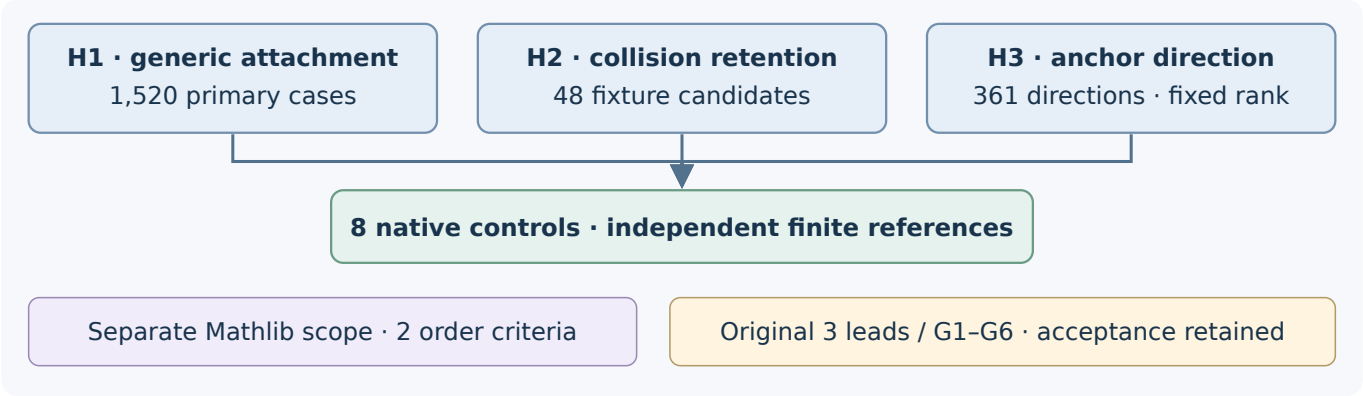
Source: `31410711806e73aed58cbc2bd74ed60b93de3572`

Eight native controls completed: six H1/H2 and two H3, zero failures/ignored. Both strict standalone clippy phases and formatting exited 0. The separate Mathlib-only project built; its scoped gate reported **2 theorems, 0 trusted exceptions**; official module kernel replay exited 0 against imports, **not** `--fresh`. These are unsigned producer observations, not independently signed execution or result standing. [pcc:unverified]

## Original requirements retained

| Lead | This cycle | Scientific obligation still open |
|---|---|---|
| Symmetric PKM | Two standard exact-order criteria. | G1–G3/G6 partial; G4/G5 unrun. Complete extraction, accounting/yield, paired external comparison, justified optimization/scaling and cold ECDLP/proof bridges. |
| Five-sum | Finite collision/anchor model. | Original frozen-session independent replay, scalar/provenance checks and complete-cost recomputation; no substitute replay. |
| Hecke base | Generic bounded-shift model. | Certified source/normalization/class labels and efficient prescribed-target attachment with qualified comparison; no Hecke generator run. |

Zero scientific panels; `comparison_admissible=false`; all cost/GAE/scaling metrics null. Full-panel pin `e99f0d598cd89d839060e991b9b157efbd6f286a` stays blocked/unfunded/unimplemented. No panel, funding, lease, IAM, quota, pool or access change; no checkpoint/protocol/runtime bypass.




Workflow diagram: fixture populations are not costs. No comparable performance values are reported.

## What H1 and H2 distinguish

**H1:** four label sets, targets 0–18 and caps 0–19 over `Z/19Z`: **1,520 primary cases**. Attachment agrees with an independent translated-set/minimum-distance reference; permutation, identity/cap and malformed-input controls are additional executions. This does not qualify production curve or Hecke backends. [pcc:unverified]

**H2:** three five-label fixtures ×16 sign vectors: **48 distinct fixture candidates**. Identity final sums are excluded. Off/on retention and repeated constructions are separate real executions, not 48 total operations. Controls check candidate/representative conservation, unchanged lookups, collision equations and independent left/right-pivot ranks. Rank, coverage and target isolation are distinct. Degenerate fixtures are not production factor bases. [pcc:unverified]

## H3: the successor cycle

Preregistered fixture: collision rank 4, query rank 5, target 5. All **361** anchors `[a,b,0,0,0]` followed the finite prediction: **342 isolate; 19 retain ambiguity**. The partition is by nonzero/zero pairing with `[1,1,1,1,1]`, not anchor count. Named independent, dependent, zero and malformed controls are included; two rank algorithms must agree. This checks coefficient row-span information with separately supplied anchor values; it does not establish their public availability in a curve problem. [pcc:unverified]

## Independent formal scope

The new Mathlib-only module contains `BinaryPrimitiveGuard.primitive_root_of_binary_guard_succ` and `BinaryPrimitiveGuard.primitive_root_of_binary_guard`: full binary power 1 and preceding half power not 1 imply exact binary order, with positive depth explicit. These specialize standard Mathlib order criteria. [pcc:unverified]

Lean 4.34.1; Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. The two declarations use only `propext`, `Classical.choice`, `Quot.sound`. No old extraction module is imported. The old stack's 34-declaration audit is separate, not this PR's two-theorem result. Rust arithmetic, representation, polynomial/extraction and complete-cost bridges remain open. [pcc:unverified]

## Evidence and retained failures

Source-bound recipe: `validation/final-v2/validate.sh` in this study. Exact local phase terminals: `/workspace/prime-leads-cycle-20261109/native-final-v2/`. Final supervisor and all native/formal phases exited 0. Initial formatting exit 1 and intermediate H3 documentation-lint exit 1 remain separate historical receipts; later success does not rewrite them. Test-run durations supply no scientific timing denominator or price. [pcc:unverified]

One canonical P20 hydration attempt with an installed CLI exited 2: no configured credentials. Required `records.jsonl` remains absent: **1,622,069 bytes**, SHA256 `2e24438edf33b233c5f06f0fc0b2f9a9dac3c55ff23eda30d7bc0cf4916e7b02`. Original replay is unperformed; synthetic outputs do not replace it. [pcc:unverified]

## Next discriminator: H4 remains unrun

Directed finite coverage may depend on label geometry beyond size alone. Freeze cap/geometry families, independent references and held-out finite sizes before execution. Separate exact finite coverage from extrapolation. H3's finite result supplies no scaling conclusion. All original scientific acceptance remains in force; software integration does not complete it.

## Sources and unchanged views

Mathlib `orderOf_eq_prime_pow` and `IsPrimitiveRoot.iff_orderOf`; this study's preregistrations/native source; original three-lead README, PKM_GOALS and PROBLEM as obligation pointers, not imported proof modules.

Canonical `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, curve and performance views stay unchanged: no admitted metric, curve/edge, phase price or reference. No speedup, timing, pricing or global-novelty conclusion is claimed. [pcc:unverified]